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Lab 07

Software Construction and Development CA

Tasks :

Task 1 – Pointer Basics

Create a program that:

- Declares an integer variable
- Creates a pointer to it
- Displays both **value and address**

Output:

(globals)

lab7.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int num = 20;
    int *ptr = &num;

    cout << "Value of num: " << num << endl;
    cout << "Address of num: " << &num << endl;
    cout << "Pointer ptr stores address: " << ptr << endl;
    cout << "Value using pointer (*ptr): " << *ptr << endl;

    return 0;
}
```

C:\Users\Danyal\Desktop\5th semester\software construction lab\lab7.exe

```
Value of num: 20
Address of num: 0x78fe14
Pointer ptr stores address: 0x78fe14
Value using pointer (*ptr): 20

--------------------------------
Process exited after 0.2528 seconds with return value 0
Press any key to continue . . .
```

Task 2 – Swap Numbers

Swap two numbers using **pointer function**.

Output:

(globals)

lab7.cpp

```cpp
#include <iostream>
using namespace std;

void swapNumbers(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    int x = 5, y = 10;

    cout << "Before swap: x = " << x << ", y = " << y << endl;
    swapNumbers(&x, &y);
    cout << "After swap: x = " << x << ", y = " << y << endl;

    return 0;
}
```

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```
Before swap: x = 5, y = 10
After swap: x = 10, y = 5

--------------------------------
Process exited after 0.2848 seconds with return value 0
Press any key to continue . . .
```

Task 3 – Dynamic Memory

Create an integer using **new** and display its value.

Output:

(globals)

lab7.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int *ptr = new int;
    *ptr = 50;

    cout << "Value: " << *ptr << endl;

    delete ptr;

    return 0;
}
```

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```
Value: 50

--------------------------------
Process exited after 0.3075 seconds with return value 0
Press any key to continue . . .
```

Task 4 – Dynamic Array

Dynamically create an array of **5 integers** and print them.

Output:

(globals)

lab7.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int *arr = new int[5];

    for (int i = 0; i < 5; i++) {
        arr[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }

    delete[] arr;

    return 0;
}
```

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```
10 20 30 40 50
--------------------------------
Process exited after 0.2776 seconds with return value 0
Press any key to continue . . .
```

esources

Task 5 – Pointer Arithmetic

Create an array and access elements using pointer arithmetic.

Output:

(globals)

lab7.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int *ptr = arr;

    for (int i = 0; i < 5; i++) {
        cout << *(ptr + i) << " ";
    }

    return 0;
}
```

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```
1 2 3 4 5
--------------------------------
Process exited after 0.3125 seconds with return value 0
Press any key to continue . . .
```